CONCEPT 49.4

Changes in synaptic connections underlie memory and learning *(pp. 1099–1101)*

- During development, more neurons and synapses form than will exist in the adult. The programmed death of neurons and elimination of synapses in embryos establish the basic structure of the nervous system. In the adult, reshaping of the nervous system can involve the loss or addition of synapses or the strengthening or weakening of signaling at synapses. This capacity for remodeling is termed **neuronal plasticity.** Our **short-term memory** relies on temporary links in the hippocampus. In **long-term memory**, these links are replaced by connections within the cerebral cortex.

? *Learning multiple languages is typically easier early in childhood than later in life. How does this fit with our understanding of neural development?*

CONCEPT 49.5

Many nervous system disorders can now be explained in molecular terms *(pp. 1102–1104)*

- **Schizophrenia**, which is characterized by hallucinations, delusions, and other symptoms, affects neuronal pathways that use dopamine as a neurotransmitter. Drugs that increase the activity of biogenic amines in the brain can be used to treat **bipolar disorder** and **major depressive disorder**. The compulsive drug use that characterizes addiction reflects altered activity of the brain's reward system, which normally provides motivation for actions that enhance survival or reproduction.
- **Alzheimer's disease** and **Parkinson's disease** are neurodegenerative and typically age related. Alzheimer's disease is a dementia in which neurofibrillary tangles and amyloid plaques form in the brain. Parkinson's disease is a motor disorder caused by the death of dopamine-secreting neurons and associated with the presence of protein aggregates.

? *The fact that both amphetamine and PCP have effects similar to the symptoms of schizophrenia suggests a potentially complex basis for this disease. Explain.*

TEST YOUR UNDERSTANDING

For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

1. Activation of the parasympathetic branch of the autonomic nervous system
(A) increases heart rate.
(B) enhances digestion.
(C) triggers release of epinephrine.
(D) causes conversion of glycogen to glucose.

2. Which of the following structures or regions is correctly paired with its function?
(A) limbic system—motor control of speech
(B) medulla oblongata—homeostatic control
(C) cerebrum—coordination of movement and balance
(D) amygdala—short-term memory

3. Patients with damage to Wernicke's area have difficulty
(A) coordinating limb movement.
(B) generating speech.
(C) recognizing faces.
(D) understanding language.

4. The cerebral cortex plays a major role in
(A) emotional memory.
(B) hand-eye coordination.
(C) circadian rhythm.
(D) breath holding.

Levels 3-4: Applying/Analyzing

5. After suffering a stroke, a patient can see objects anywhere in front of him but pays attention only to objects in his right field of vision. When asked to describe these objects, he has difficulty judging their size and distance. What part of the brain was likely damaged by the stroke?
(A) the left frontal lobe
(B) the right frontal lobe
(C) the right parietal lobe
(D) the corpus callosum

6. Injury localized to the hypothalamus would most likely disrupt
(A) regulation of body temperature.
(B) short-term memory.
(C) executive functions, such as decision making.
(D) sorting of sensory information.

7. **DRAW IT** The reflex that pulls your hand away when you prick your finger on a sharp object relies on a neuronal circuit with two synapses in the spinal cord. (a) Using a circle to represent a cross section of the spinal cord, draw the circuit. Label the types of neurons, the direction of information flow in each, and the locations of synapses. (b) Draw a simple diagram of the brain indicating where pain would eventually be perceived.

Levels 5-6: Evaluating/Creating

8. **EVOLUTION CONNECTION** Scientists often use measures of "higher-order thinking" to assess intelligence in other animals. For example, birds are judged to have sophisticated thought processes because they can use tools and make use of abstract concepts. Identify problems you see in defining intelligence in these ways.

9. **SCIENTIFIC INQUIRY** Consider an individual who had been fluent in American Sign Language before suffering an injury to his left cerebral hemisphere. After the injury, he could still understand that sign language but could not readily generate sign language that represented his thoughts. Propose *two* hypotheses that could explain this finding. How might you distinguish between them?

10. **SCIENCE, TECHNOLOGY, AND SOCIETY** With increasingly sophisticated methods for scanning brain activity, scientists are developing the ability to detect an individual's particular emotions and thought processes from outside the body. What benefits and problems do you envision when such technology becomes readily available? Explain.

11. **WRITE ABOUT A THEME: INFORMATION** In a short essay (100–150 words), explain how specification of the adult nervous system by the genome is incomplete.

12. **SYNTHESIZE YOUR KNOWLEDGE**

Imagine you are standing at a microphone in front of a crowd. Checking your notes, you begin speaking. Using the information in this chapter, describe the series of events in particular regions of the brain that enabled you to say the very first word.

For selected answers, see Appendix A.

Explore Scientific Papers with Science in the Classroom AAAS
How does a neuron develop different functions than those of similar neighboring neurons?
Go to "New Neurons Stand Out from the Crowd" at www.scienceintheclassroom.org.
Instructors: Questions can be assigned in Mastering Biology.